Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_best_single_fixed_heuristic, phase8_full_solver_evolution, phase8_random_portfolio.

Run Metadata

- run_name: run_20260520_220319_n
- started_at_local: 2026-05-20 22:03:19
- finished_at_local: 2026-05-20 22:24:00
- duration_hhmm: 00:21
- duration_seconds: 1240.802
- seed_offset: 13000
- replicate_label: n
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	694.051829	0.07226	0.007284	0.031222	0.0	0.0	True
phase8_random_portfolio	random_portfolio	none	0.20534	0.13308	112.013529	0.20534	0.051131	0.107945	0.0	0.0	True
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	703.521671	0.073246	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	704.132457	0.07331	0.007284	0.031222	0.0	0.0	False
phase8_llm_static_selector	static_selector	none	0.214311	0.142051	226.7406	0.214311	0.024668	0.094536	0.0	0.62	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.086887	0.014627	1331.502514	0.166688	0.003547	0.034251	0.65746	0.7275	False
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.120261	0.048001	5320.915529	0.921975	0.090132	0.101232	0.754937	0.695	False
phase8_full_solver_evolution	full_solver	none	0.213387	0.141127	99.356914	0.213387	0.150269	0.173523	0.893412	0.56	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 694.051829 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.20534, selector regret 0.13308, runtime-adjusted gap 0.20534.
- Family transfer 0.051131 and combined transfer 0.107945.
- Runtime 112.013529 ms, runtime inflation -0.838609, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.073246.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 703.521671 ms, runtime inflation 0.013644, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07331.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 704.132457 ms, runtime inflation 0.014524, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.024668 and combined transfer 0.094536.
- Runtime 226.7406 ms, runtime inflation -0.673309, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:farthest_insertion_two_opt:annealed_multi_start:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": 2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selection over a frozen portfolio using geometry/structure descriptors; uses bounded schedule tuning parameters for stagnation and perturbation intensity.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.08, "name": "tsp_heuristic_oracle_portfolio_static_v1", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.15, "time_budget_trigger": 0.65, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.086887, selector regret 0.014627, runtime-adjusted gap 0.166688.
- Family transfer 0.003547 and combined transfer 0.034251.
- Runtime 1331.502514 ms, runtime inflation 0.918448, mean novelty 0.65746, and complexity 0.7275.
- Pareto efficient: False.
- Controller signature: annealed_multi_start:cluster_first_local_search:farthest_insertion_two_opt:annealed_multi_start:stag3:cand0:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": 0, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "edge_preserving_restart", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selector over a frozen heuristic portfolio. Chooses construction/refinement style from interpretable structural descriptors (clustered/grid/two-cluster-bottleneck/corridor/nearest-neighbor trap) and escalates to restart/annealin", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold":

0.12, "name": "tsp_hh_frozen_portfolio_instance_adaptive_v3", "nearest_neighbor_trap_heuristic": "farthest_insertion_two_opt", "random_like_heuristic": "annealed_multi_start", "restart_offset": 1, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.12, "time_budget_trigger": 0.6, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.120261, selector regret 0.048001, runtime-adjusted gap 0.921975.
- Family transfer 0.090132 and combined transfer 0.101232.
- Runtime 5320.915529 ms, runtime inflation 6.666453, mean novelty 0.754937, and complexity 0.695.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:annealed_multi_start:stag3:cand-1:restart2.
- Controller rule summary: {"acceptance_bias": 0.008, "candidate_limit_offset": -1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selection over a frozen TSP heuristic portfolio with bounded, interpretable switches driven by stagnation and perturbation failure.", "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.12, "name": "interp_portfolio_adaptive_tsp_v2", "nearest_neighbor_trap_heuristic": "cheapest_insertion_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 2, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.05, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.213387, selector regret 0.141127, runtime-adjusted gap 0.213387.
- Family transfer 0.150269 and combined transfer 0.173523.
- Runtime 99.356914 ms, runtime inflation -0.856845, mean novelty 0.893412, and complexity 0.56.
- Pareto efficient: True.

Judge Appendix

Conservative Interpretation of TSP Heuristic Portfolio Suite Results

Condition	Held-out TSPLIB Gap	Selector Regret	Runtime-Adjusted Gap	Runtime Inflation	Transfer Gap	Pareto Efficient	Notes
phase8_best_single_fixed_heuristic	**0.07226**	0.0	0.07226	0.0	0.031222	Yes	Best held-out TSPLIB gap, zero regret (fixed/frozen heuristic farthest_insertion_two_opt); baseline for adaptive selectors
phase8_oracle_selector	0.07226	0.0	0.073246	0.0136	0.026678	No	Oracle selector matches best fixed heuristic gap, slight runtime inflation; upper bound, not deployable

Condition	Held-out TSPLIB Gap	Selector Regret	Runtime-Adjusted Gap	Runtime Inflation	Transfer Gap	Pareto Efficient	Notes
phase8_supervised_ml_selector	0.07226	0.0	0.07331	0.0145	0.031222	No	Nearly identical performance to best fixed heuristic and oracle; supports interpretable instance-adaptive control
phase8_llm_evolved_adaptive_controller	0.086887	0.014627	0.166688	0.9184	0.034251	No	Adaptive control shows some selector regret and runtime inflation (~x2); worse held-out gap and runtime-adjusted gap
phase8_llm_static_selector	0.214311	0.142051	0.214311	-0.6733	0.094536	No	Static selectors have worse gaps and regret despite runtime saving
phase8_random_portfolio	0.20534	0.13308	0.20534	-0.8386	0.107945	Yes	Random portfolio shows poor gaps but low runtime, Pareto efficient due to runtime trade-off
phase8_full_solver_evolution	0.213387	0.141127	0.213387	-0.8568	0.173523	Yes	Full solver evolution underperforms in gap but is Pareto efficient due to low runtime
phase8_llm_evolved_controller_diversity_failure_replay	0.120261	0.048001	0.921975	6.6664	0.101232	No	Very high runtime inflation (~7x), much worse runtime-adjusted gap; regret and transfer gap also high

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Key Findings and Recommendations

- **Primary Endpoint (Held-out TSPLIB Gap):**
- The **best single fixed heuristic (farthest_insertion_two_opt)** achieves the lowest gap (0.07226), matching the oracle selector.
- Adaptive controllers, including ML-based and LLM-evolved selectors, do **not improve held-out gap** beyond best fixed.
- Some adaptive methods (e.g., phase8_llm_evolved_adaptive_controller) actually degrade held-out gap, with significant runtime inflation.

- **Selector Regret vs Oracle:**
- The oracle selector sets a tight lower bound on regret (0.0).
- The best static and supervised ML selectors achieve zero regret (no gap from oracle).
- Adaptive controllers incur small but noticeable regret (1.5% to 5%), suggesting imperfect selection accuracy or adaptation.

- **Runtime-Adjusted Gap and Runtime Inflation:**
- The best fixed heuristic and oracle have comparable low runtime-adjusted gaps (~0.07).
- Adaptive controllers often incur significant runtime inflation (close to doubling or more), resulting in worse runtime-adjusted gaps (~0.17 and higher).

- Static selectors reduce runtime (negative inflation) but pay with worse solution quality and regret.
- The large runtime overhead in some adaptive controllers limits practical interpretability and deployment.

- **Pareto Efficiency:**
- Only three conditions are Pareto efficient: best fixed heuristic, random portfolio, and full solver evolution.
- Neither adaptive selector variants reach Pareto front, indicating suboptimal tradeoffs between gap and runtime.

- **Cross-Family Transfer (Transfer Gap):**
- Transfer gaps are generally low for best fixed, oracle, and supervised selectors (~0.03).
- Larger transfer gaps appear with adaptive selectors and full solver evolution, indicating less robust cross-family generalization.

- **Interpretability and Control:**
- Static and supervised ML selectors demonstrate close-to-oracle performance with minimal runtime inflation, supporting claims for interpretable instance-adaptive control.
- Evolved adaptive controllers demonstrate promising adaptation efficiency but suffer from high runtime overhead, limiting practical benefit.
- No evidence of discovering new heuristics; results reflect hyper-heuristic control over known methods.

Summary

- The **best single fixed heuristic (farthest_insertion_two_opt)** remains the strongest baseline in held-out TSPLIB gap and runtime-adjusted performance.
- The **oracle selector confirms this baseline as a tight lower bound**, so adaptive improvement is challenging.
- **Supervised ML and static selectors nearly match the oracle without significant runtime cost**, supporting interpretable, deployable instance-adaptive control.
- Evolved adaptive controllers show limited benefit due to **runtime inflation and increased regret**, diminishing practical value.
- The portfolio suite should emphasize **maintaining interpretability and runtime efficiency** as best fixed heuristics and simpler selectors demonstrate strong, reliable performance.